

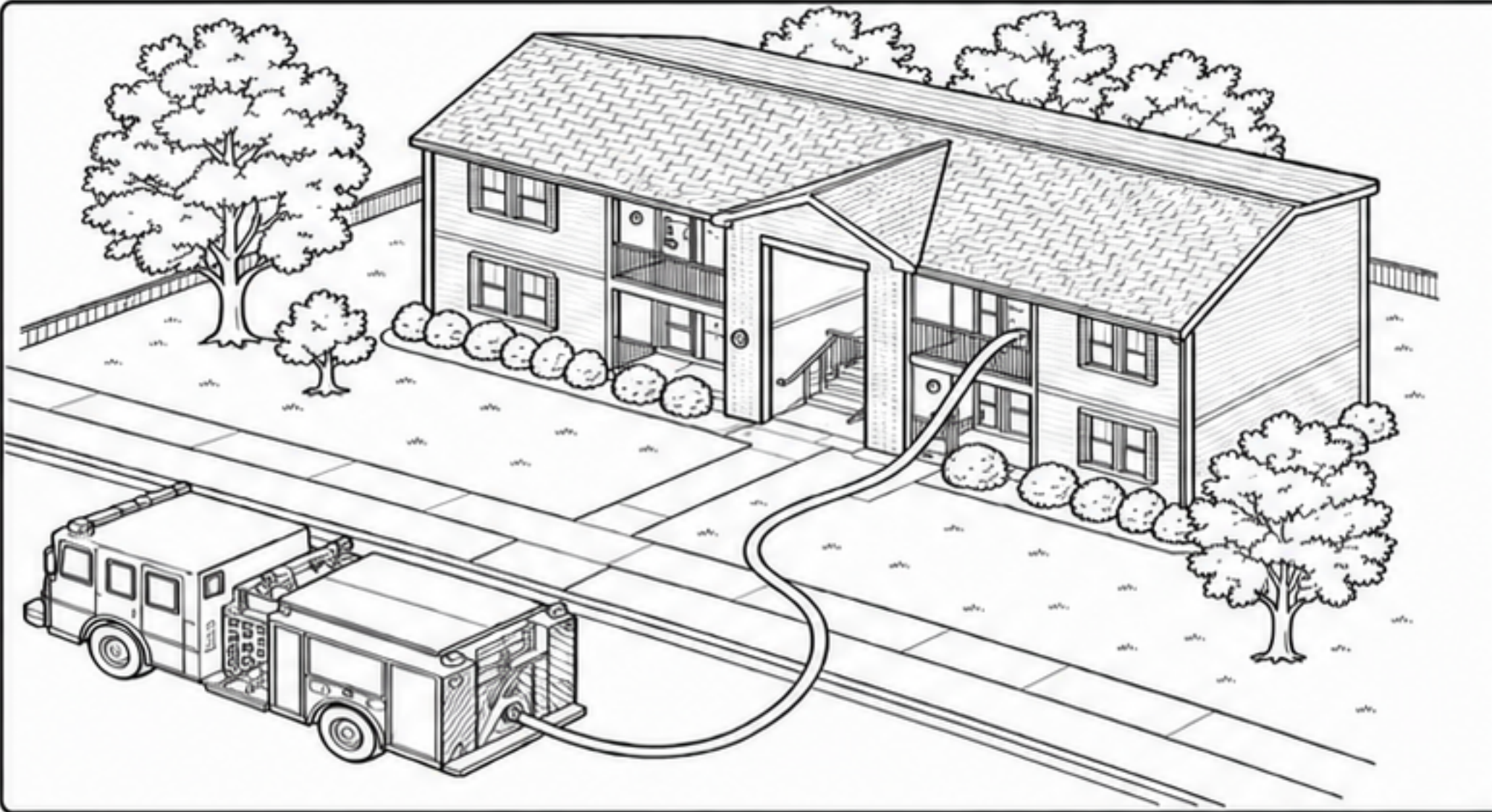
FirePumpSim Printable Scenario

Driver/Operator Practice Worksheet

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Student Name: _____ Date: _____ Class: _____ Score: _____

Scenario: Apartment Fire - Single Attack Line



Given / Reference

- Apparatus: Engine 181
- Water source: Hydrant supply established
- Attack line: 200 ft of 1¾-inch hose
- Nozzle: Fog nozzle, 150 GPM at 100 PSI
- Hose coefficient (C): 15.5
- Appliance loss: 0 PSI
- Elevation: 2nd floor (+5 PSI)
- Estimated pump-to-entry travel: included in the 200 ft hose length

Student Questions

1. Using $FL = C \times Q^2 \times L$, what is the friction loss in the 200-foot 1¾-inch attack line?

2. What pump discharge pressure (PDP) is required to supply this line to the 2nd-floor apartment entrance?

3. If a second 200-foot 1¾-inch line with the same nozzle is stretched to the 1st-floor apartment entrance, what would the total fire flow in GPM?

4. If both lines are flowing, what friction loss would each 200-foot line have?

5. What pump discharge pressure (PDP) would be required to supply the 1st-floor line described in Question 3?

Required Equations

- $FL = C \times Q^2 \times L$
- $PDP = NP + FL + \text{Appliance Loss} + \text{Elevation}$
- $Q = GPM / 100$
- $\text{Total Flow} = \text{Sum of All Operating Line Flows}$

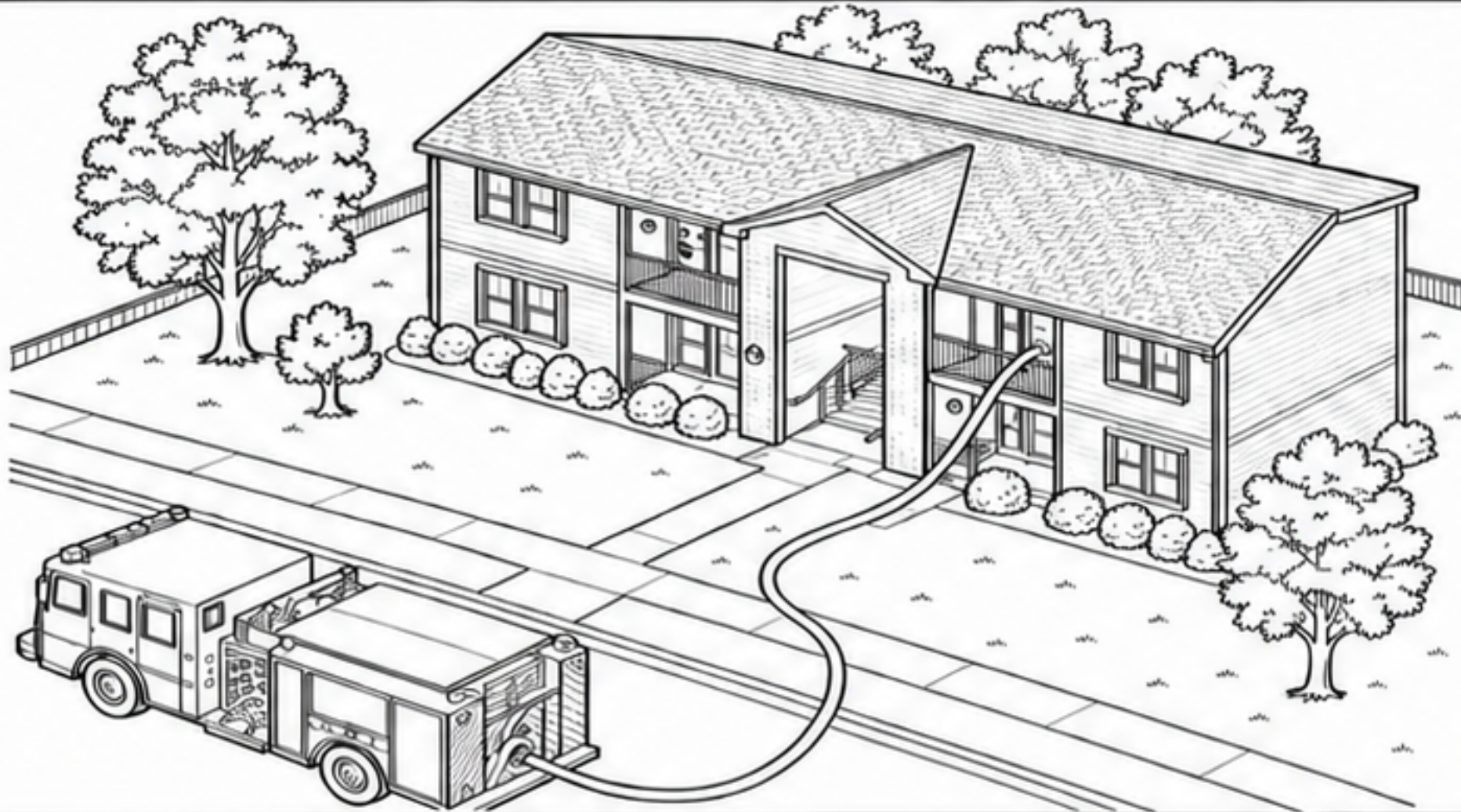
FirePumpSim Answer Key

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Scenario: Apartment Fire - Single Attack Line



Scenario Givens

- Apparatus: Engine 181
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- Attack line: 200 ft of 1¾-inch hose
- Nozzle: Fog nozzle, 150 GPM at 100 PSI
- Hose coefficient (C): 15.5
- Appliance loss: 0 PSI
- Elevation: 2nd floor (+5 PSI)
- Estimated pump-to-entry travel: included in the 200 ft hose length

Instructor Answer Key

1. Friction loss in the 200-foot 1¾-inch attack line	Formula: $FL = C \times Q^2 \times L$ $Q = 150 / 100 = 1.5$ $L = 200 / 100 = 2$	$FL = 15.5 \times (1.5)^2 \times 2$ $FL = 15.5 \times 2.25 \times 2$ $FL = 69.75 \text{ PSI}$	Rounded Answer: 70 PSI
2. Required pump discharge pressure (PDP)	Formula: $PDP = NP + FL + \text{Appliance Loss} + \text{Elevation}$ $PDP = 100 + 70 + 0 + 5$		Answer: 175 PSI
3. Total fire flow with a second line to the 1st floor	Each line flow = 150 GPM Total Flow = 150 + 150		Answer: 300 GPM
4. Friction loss if both lines are flowing	Because the two lines are identical and calculated separately, the friction loss in each 200-foot line remains the same.		Answer: 70 PSI friction loss in each line
5. Required PDP for the 1st-floor line	Formula: $PDP = NP + FL + \text{Appliance Loss} + \text{Elevation}$ $PDP = 100 + 70 + 0 + 0$		Answer: 170 PSI

Accept reasonable rounding: ± 1 PSI.